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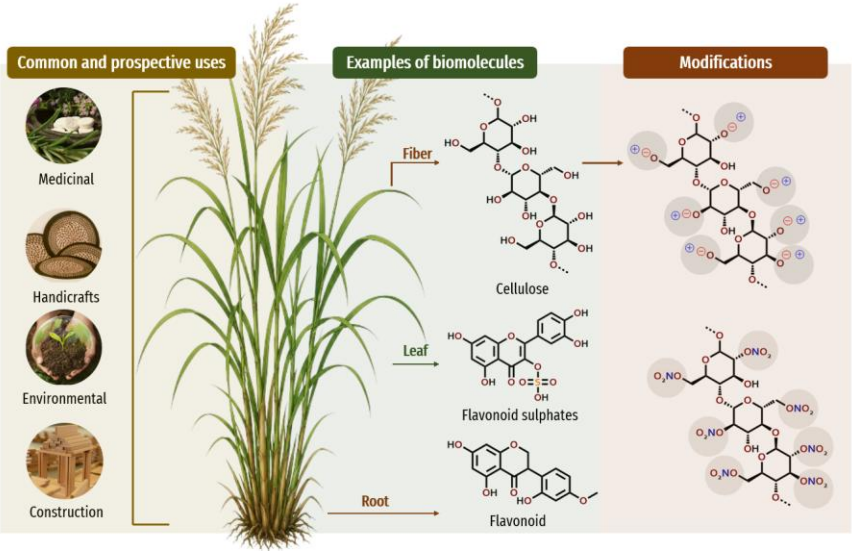
An overview of Gynerium sagittatum – Part II: Chemical characteristics, common and prospective uses

Nathaly Sánchez-Zuleta [1], Angélica García-Quintero [1,2], Manuel Palencia [2]

[1] *Mindtech Research Group (Mindtech-RG), Mindtech s.a.s., Cali – Colombia.*
[2] *Research Group in Science with Technological Applications (GI-CAT), Department of Chemistry, Faculty of Natural and Exact Sciences, Universidad del Valle, Cali – Colombia.*

Corresponding Author: Manuel Palencia. E-mail: manuel.palencia@correounivalle.edu.co

Graphical Abstract



Abstract. *Gynerium sagittatum*, commonly known as Arrow Cane, is an herbaceous plant primarily found in Latin America, ranging from southwestern Mexico to northern Brazil and Paraguay. In these regions, it is not cultivated on an industrial scale and is primarily harvested for the cellulose fibers in its leaf veins. These fibers are then woven into decorative and practical items, such as baskets, hats, and fences. Despite these relevant uses, the technological advancement associated with the arrow cane remains limited. There is minimal research and planning regarding the utilization of other parts of the plant, its extracts, and the modification of its fibers for bio-based and multifunctional materials. Therefore, it is essential to consolidate the available information on the chemical properties, recognized uses, and potential applications of this crop to better understand the plant and encourage research into its utilization. In this sense, the second part of this review focuses on the limited chemical composition reported for the various organs of the plant, it discusses common uses that extend beyond handicrafts to include ethnomedical applications, structural materials, and environmental remediation, additionally, it explores potential uses of the fiber through its valorization via chemical and physical modifications to disseminate existing knowledge on this topic and encourage the scientific community to investigate about this plant and its promising applications.

Keywords: Arrow Cane, Cellulosic fiber, Botanical medicine, Flavonoids, Mercerization, Acetylation.

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1. Introduction

Gynerium sagittatum is a woody grass-like plant that grows in tropical and subtropical regions, where it is valued primarily for its cultural and traditional significance; its connection to productive activities is associated almost exclusively with the production of handicrafts, particularly in Colombia for the making of the *Vueltiao hat*, an iconic symbol of the nation's culture (Casas Caro et al., 2010; 2018; Ramos, 2019; Royal Botanic Gardens, 2022). Given this scenario, knowledge about this plant has been predominantly empirical, derived from observations and cultural traditions that have focused particularly on the plant fiber used in the production of handicrafts; likewise, research on *Gynerium sagittatum* has advanced and has been conducted primarily in the areas of agronomy, biotechnology for propagation, and the utilization of the fiber (Ramos, 2019; Rojas Barbosa, 2003).

Therefore, there is a knowledge gap regarding the limited information and physicochemical characterization of the *Martinera* and *Costera* ecotypes of this plant and, more generally, about parts of the plant other than the root and fiber, such as the stem, reproductive structures, fruits, and seeds (Jaimes Roncancio, 2023). Additionally, there is a lack of technical information on other recognized applications of this plant, including its therapeutic properties, its potential as a bioaccumulator for environmental remediation, and the modification of plant fiber to enhance its use in structural and functional materials and systems.

To address this information gap, it is imperative to understand the physicochemical characteristics of the plant's components, which permits the establishment of the possibilities and limitations of these structures, as well as an understanding of their traditional and prospective uses. Based on this argument, the second part of this review addresses the limited chemical composition data reported on the plant's various organs, its customary uses - which include not only handicrafts but also ethnomedical applications, structural materials, and environmental remediation through phytoaccumulation- as well as the potential uses of the fiber through its valorization via chemical and physical modifications. This overview aims to contribute to a comprehensive understanding and the sustainable use of this crop, as well as to the technological development of functional materials based on its fibers, and to encourage research and development regarding the plant's potential therapeutic uses, the development of structural materials and composites, and the utilization of waste derived from handicraft production.

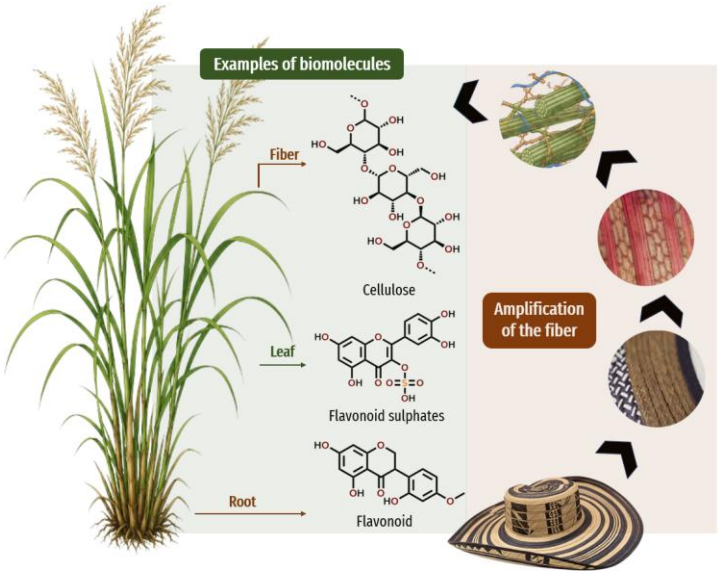


Figure 1. Certain biomolecules identified in the plant organs and fiber of *Gynerium sagittatum*, along with the macroscopic and microscopic appearance of the fiber.

2. Chemical composition

Gynerium sagittatum is classified as a "primary plant" based on its natural fiber content, that is, a plant grown for its fiber, since this is its primary use and source of commercial and cultural interest (Dominguez Huarcaya et al., 2023; Rowell, 2008). Therefore, after the plant is harvested and the fiber is extracted, the resulting biomass is considered waste despite potentially containing biomolecules of interest, a situation that also occurs to a large extent with other plant organs once they reach the end of their useful life (see **Figure 1**). This section therefore aims to highlight the importance and the available information regarding the chemical composition of the plant organs of the Arrow Cane plant, in order to understand its applications and scientific potential, and the valorization opportunities for improving the quality of life for communities that rely on this plant and its processing as an economic activity and source of livelihood.

In the case of *vegetable fiber*, its proximate composition, particularly that of a sample from the Colombian Caribbean region, showed a moisture content of 1.13 %, an ash content of 2.87 % at 600 °C, a protein content of 5.17 %, a carbohydrate content of 20.95 %, and a cellulose content of 68.58 %, which is characteristic of a cellulosic fiber, composition that confers flexibility properties to the material (Programa Nacional de Conformación de Cadenas Produc-

tivas para el Sector Artesanal, 2005). This fiber contains not only cellulose but also a mixture of structural polysaccharide compounds: lignin, cellulose, and hemicellulose. Lignin is an aromatic biopolymer with no fixed structure; it is formed by the arbitrary linking of phenolic alcohol units, known as monolignols; cellulose is a polysaccharide composed of linear chains of β-D-glucose molecules connected by β(1→4) glycosidic bonds; and hemicellulose is an amorphous and branched heteropolysaccharide made up of a main chain of five- and six-carbon sugars, which have diverse substituent groups (Franca et al., 2019). These polysaccharides account for approximately 91.6 % w/w, while non-structural components, such as waxes, proteins, and fats, account for 8.43 % w/w of Arrow Cane fiber (Franca et al., 2019).

Furthermore, these polysaccharides present an associated structure in which cellulose microstructures are connected by amorphous lignin and hemicellulose (de Lima et al., 2022; Kommula et al., 2014). The bonding interactions in crystalline cellulose are primarily based on a molecular organization of the cell wall (see **Figure 1**), which is mediated by three main types of hydrogen bonds. These are: *Intrachain bonds*, formed between hydroxyl groups from neighboring cellulose molecules within the same chain. *Interchain bonds*, which occur between cellulose chains within the same sheet. *Intersheet bonds* are formed between staggered chains of different sheets (Dong et al., 2016). This structural organization allows for recalcitrance, defined as the resistance of cell walls to degradation; furthermore, this chemical composition and organization confer hydrophobic properties on the fiber (Dong et al., 2016; Programa Nacional de Conformación de Cadenas Productivas para el Sector Artesanal, 2005). In turn, the crystalline domains of the fiber are composed of cellulose, while the disordered regions are associated with lignin and hemicellulose. The degree of crystallinity varies based on factors such as the type of fiber, its origin, and the conditions under which it was formed. Additionally, reactions that modify the hydroxyl groups in the cellulose structure can decrease the fiber's crystallinity by introducing new groups that alter the material's structure (Marvila et al., 2021; Mayer-Laigle et al., 2020; Poletto et al., 2014; Rasheed et al., 2020; Sena Neto et al., 2013; Thygesen et al., 2005).

Regarding the *stem*, it also contains cellulose, hemicellulose, and lignin; however, unlike the fiber, its lignin content is higher, which makes it stiffer and more brittle. Furthermore, research has shown that the methanolic extract of young stem plant material contains a moderate amount of phenolic compounds and anthraquinones, an abundant amount of free amino acids, and a much higher proportion of flavonoids in its phytochemical profile (Dominguez Huarcaya et al., 2023).

Concerning the *leaves* from which the fiber is obtained, the presence of flavonoid sulfates has been reported; these help regulate secondary metabolism, chemical defense, and growth, and are usually sulfate esters of common hydroxyflavones and hydroxyflavonols and their methyl esters (Barron et al., 1988; Programa Nacional de Conformación de Cadenas Productivas para el Sector Artesanal, 2005; Watson et al., 2023). Nevertheless, the most extensively studied source of flavonoids in this plant is the root, which has been shown to contain high levels of biologically active flavonoids, and especially isoflavonoids, making it an important source of phytoestrogens, which have been reported to have high potential as antioxidants, with beneficial effects on human and animal health by acting as therapeutic biomolecules against diseases such as cancer by inhibiting the proliferation of cancer cells and stimulating the immune system (Benavides et al., 2007; Cayetano-Salazar et al., 2021; Noriega-Rodríguez et al., 2020; Roumy et al., 2020). **Figure 2** shows the main flavonoids identified in the methanolic and chloroform extracts of dried roots of *Gynerium sagittatum* from Peru, where molecules such as biochanin A and dalbergioidin can reach significant concentrations ranging from 373 ± 21 to 163 ± 3 ppm, respectively (Benavides et al., 2007). In turn, other chemical compounds have been identified in these extracts, such as *Z*-cinnamic acid, *E*-cinnamic acid, *p*-coumaric acid, and ferulic acid from the phenylpropanoid family, as well as 2-propenoic acid from the carboxylic acid family (Benavides et al., 2007).

The proportion of the aforementioned phytochemicals also depends on soil and climate conditions, as well as the plant's age and ecotype. In turn, the mechanical properties of the fiber - such as density, length, diameter, elongation at break, and tensile strength - are also influenced by factors such as the region, the plant's nutritional status and health, its age, and its variety. Furthermore, the lack of information on the chemical composition of the seeds, fruit, and panicles of *Gynerium sagittatum* is noticeable, presenting a significant opportunity for research in ethnobotany.

3. Recognized applications

3.1. Handicrafts

Ritual objects made from arrow cane leaves have been found at a temple on the northern coast of Peru, near the city of Trujillo, and are believed to have been burned during purification rituals for warriors of the Mochica culture, which dates from the 1st to the 8th centuries A.D. (Jaimes Roncancio, 2023). In terms of aesthetic aspects, the panicle of *Gynerium sagittatum* has been used as a home decor item, primarily as a vase, both in its natural state and dyed in bright colors; in turn, the rachis that supports the inflorescence has been used to make crafts with bird feathers (Jaimes Roncancio, 2023). Furthermore, the most significant handicrafts are made by braiding fibers obtained from the veins of the plant's leaves, which are used to create accessories such as bags and necklaces, mats, baskets, hats, and more than fifty types of handicrafts produced primarily in Tuchín and San Andrés de Sotavento in the Sabana de Córdoba region of Colombia (Rojas Barbosa, 2003). Whereas the Vueltiao hat (see **Figure 1**) is the product of greatest economic (it is also one of the most widely sold artisanal items in the country) and cultural significance in Colombia - with designs associated with artistic, religious, and political and social expressions - typical of the country's northern coast and primarily produced by the Zenú indigenous peoples and their descendants (iNaturalist, 2022; Martínez Osorio et al., 2020; Pastrana & Suárez Padrón, 2009).

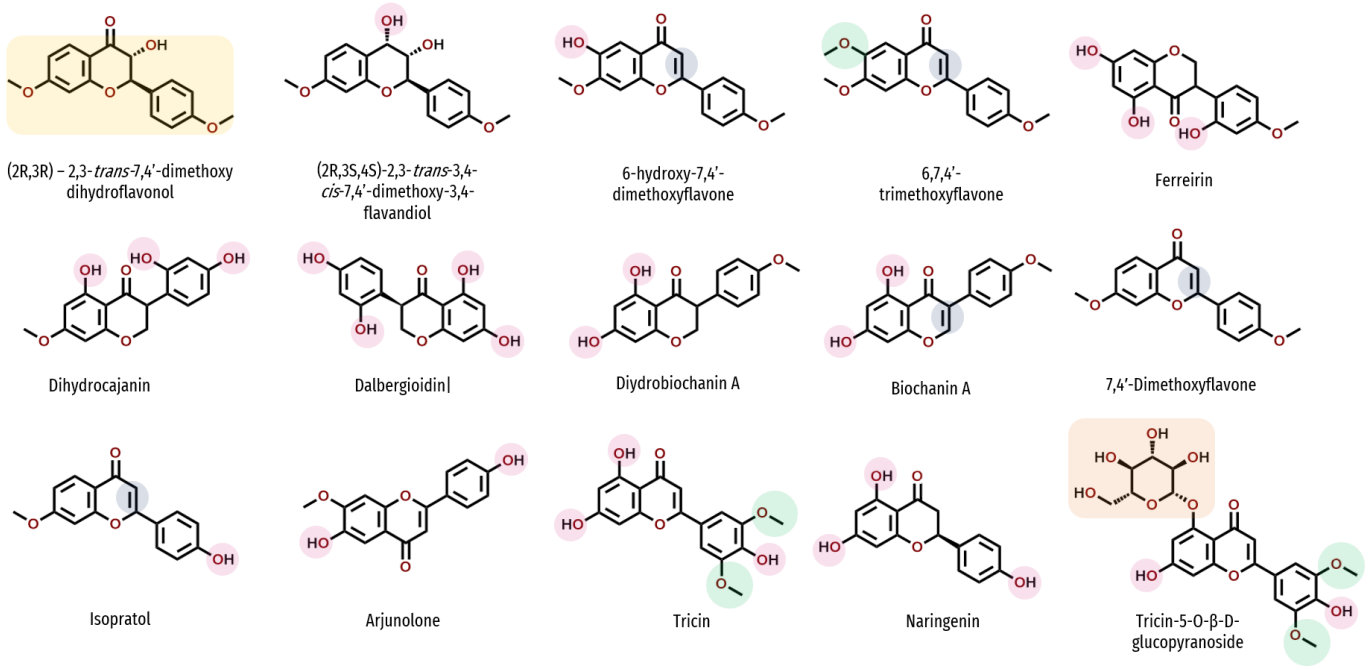


Figure 3. Main flavonoid compounds extracted from the dried roots of *Gynerium sagittatum*.

The Vueltiao hat is typically produced through a series of steps: First, the leaves of the Criolla variety are harvested, and the midrib is removed. Next, the outer skin or residue of the leaf is stripped away to obtain clean fibers, using a medium-edged knife while ensuring the artisan's leg is protected. After scraping, the fibers are split lengthwise to create narrower strips called *ripias*, which are 1 to 2 mm wide and 1.2 to 1.7 cm long. These strips are sorted into clean fibers and pigmented fibers; the clean fibers are used for bleaching, while the pigmented fibers are used for dyeing, especially to produce black fibers. For the bleaching process, the arrow cane fibers are submerged in sour cane water and boiled overnight with bleaching agents like citric acid, sodium hypochlorite, or hydrogen peroxide. After boiling, they are washed and exposed to sunlight. The pigmented fibers are dyed using a combination of alkaline clay and plant materials at an alkaline pH. The fibers are left in this solution for several days before being washed. Following this, they are heated with dye plants such as annatto (*Bixa orellana*) or toothpaste (*Pithecellobium dulce*) for about two hours. This dyeing procedure can be repeated two or three times to achieve a deeper shade of the desired color. Once the bleached and dyed fibers are prepared, they are braided in pairs or bundles. The quality of the braid depends on the number of pairs used; documented braids include 7, 9, 11, 15, 19, 21, 23, 25, 27, and 31 pairs, with those using the highest number generally being the finest. After producing the required length of braid, the hat is formed using a template that incorporates 5 to 7 wraps. The braid then changes direction to form the hat's crown. Once the crown is completed, the braid changes direction again and extends outward to create the brim. Finally, the edge of the brim is finished with a trim, and the braids are joined together by stitching (Rojas Barbosa, 2003; Suárez et al., 2017).

3.2. Medicinal use

The medicinal use of *Gynerium sagittatum* has primarily been studied and utilized within artisanal practices and traditional folk medicine. This plant is generally reported to be effective in treating a variety of conditions, including varicose ulcers, osteoporosis, intestinal parasites, liver diseases, topical ulcers, fungal infections, as well as female infertility and prostate diseases (Benavidez et al., 2007; Desmarchelier et al., 1996; Muñoz et al., 2000; Roumy et al., 2020; Svetaz et al., 2010; Teisher and Stimmel, 2022). Regarding the latter condition, specifically, the therapeutic potential of the Arrow Cane plant for the treatment of benign prostatic hyperplasia has been reported. Scientific studies in this area have shown that the *root* extract of *Gynerium sagittatum* has a beneficial effect, using *Mus musculus* as an animal model, associated with the inhibition of the enzymatic process that converts testosterone into dihydrotestosterone, the primary androgen responsible for stimulating prostate growth, resulting in a reduction in inflammation and the growth of this gland (Dominguez Huarcaya et al., 2023). While infusions made from the *root* have been recognized for their ability to stop hair loss, as a diuretic, as a pain reliever for colic in cattle and horses, and as a treatment for snake bites (SiB Colombia, 2022, Giovannini & Howes, 2017; Hoyos Pajares et al., 2011; iNaturalist, 2022; Rojas Barbosa, 2003).

Also, among the chemical compounds extracted from the roots, ferreirin, dihydrocajanin, dalbergioidin, and dihydrobiochanin have been shown to possess antibacterial activity against cariogenic bacteria (Benavidez et al., 2007). Biochanin A exhibited estrogenic, anti-inflammatory, and anticancer activity in in vitro and in vivo assays (Benavidez et al., 2007).

On the other hand, *stem* extracts have been used as empirical treatments with anti-inflammatory, analgesic, and diuretic properties (Benavidez et al., 2007; Dominguez Huarcaya et al., 2023). In addition, a product called "bud juice" is extracted from the inner part of young stems and has been used to treat acne, urinary retention, and earaches (Rojas Barbosa, 2003; Teisher & Stimmel, 2022). It is also important to highlight that the antimalarial potential of stem extracts of Arrow Cane has been evaluated both *in vitro* against chloroquine-resistant *Plasmodium falciparum* strains *and in vivo* against a rodent malaria model using *Plasmodium vinckei petteri*, yielding an inhibition percentage of 100 % at a concentration of 10 micrograms/mL and 11 % at a dose of 522 mg/kg over 4 days, respectively, which indicates excellent *in vitro* inhibition and no activity in the *in vivo* assay (Muñoz et al., 2000).

As for the *leaf* extracts, they have been used to treat asthma, eye pain, diarrhea, prostatitis, depression, body aches, loss of appetite, snake bites, urinary tract inflammation, burns, fungal infections, wounds, and anemia (Bourdy et al., 2000; Desmarchelier et al., 1996; Dominguez Huarcaya et al., 2023; Jaimes Roncancio, 2023; Ramos, 2009; Valadeau et al., 2010). As for the *bud*, it has been used to treat skin conditions, as a topical analgesic, and in small doses to treat whooping cough (Casas Caro, 2010; Rojas Barbosa, 2003; Teisher & Stimmel, 2022). Furthermore, macerated and boiled shoots have been recognized as an analgesic for kidney pain (Rojas Barbosa, 2003).

These medicinal properties and applications indicate the presence of bioactive metabolites with therapeutic potential; therefore, it is of interest to investigate the plant's phytochemical composition in greater depth, considering the plant organ, stage of maturity, ecotype, and its response to biological assays. For example, toxicity against *Artemia salina* has been investigated, with the aqueous preparation exhibiting an LC_{50} of 2,474 µg/mL and the ethanolic preparation having an LC_{50} >10,000 µg/mL, which corresponds to relatively low toxicity for this animal model; this is a relevant assay that encourages further research into the biological activity of this plant and its therapeutic use (Bussmann et al., 2011).

3.3. Structural applications

After the plant blooms, its stem is used to build walls, wattle-and-daub houses, kiosks, and ceilings; to fence homes; to create laminates and panels; and to construct pens and sheds (Casas Caro, 2010; Jaimes Roncancio, 2023; Rivera Jiménez et al., 2009; Rojas Barbosa, 2003; Suárez Padrón et al., 2009; Teisher & Stimmel, 2022).

In this sense, despite the structural strength of the *Gynerium sagittatum* stem, surface modifications have been evaluated to improve the structural properties of other plant organs and products derived from this plant. For example, the fibers have been modified with castor oil-based polyurethane resin to improve their surface rigidity, which is the first step toward facilitating impregnation for the production of compact panels for construction; as for a rice husk model, it has been pretreated using the steam explosion technique at a pressure of 2 MPa to remove structural materials such as lignin and hemicellulose and enhance the physical properties of the final material; in turn, the incorporation of Kraft lignin improved the panel's mechanical properties (Diossa et al., 2017; Monsalve Alarcón et al., 2018; Naveda Rengifo et al., 2019).

3.4. Environmental remediation

Gynerium sagittatum has been planted to protect the integrity and structure of riverbanks, as its rhizomes grow approximately 1 meter below and/or at ground level in a longitudinal direction, thereby helping to control erosion (SiB Colombia, 2022). Furthermore, it has been shown that *Gynerium sagittatum* can bioaccumulate heavy metals such as Cd(II), Hg(II), Cr(VI), and Pb(II) (Madera-Parra et al., 2015; Ortega-Ortega et al., 2011). In particular, a study was conducted to evaluate the phytoremediation potential of Arrow Cane in constructed wetlands treating synthetic landfill leachate. The plant exhibited healthy growth over 60 days in this medium containing heavy metals. Cadmium was detected in all of the plant's organs, with a root-to-shoot ratio greater than 1, indicating that the roots had a higher accumulation capacity for this metal compared to the shoots, which suggests a low mobility of cadmium to other plant tissues. For chromium, the translocation factor was less than 1, indicating a slight decrease with increasing metal concentration in the water. The study concluded that *Gynerium sagittatum* was the most effective species for phytoaccumulating the four tested metals, followed by *Heliconia psittacorum* and *Colocasia esculenta*. These findings demonstrate that constructed wetlands with Arrow Cane can be an effective technology for treating landfill leachate on a larger scale (Madera-Parra et al., 2015).

Furthermore, *Gynerium sagittatum* fiber has been evaluated as a potential material for the environmental remediation of oil-contaminated water through various modifications. The fiber was modified using two different treatments: *i)* treatment with the protic ionic liquid, 2-hydroxyethanolamine acetate, and *ii)* a combination of mercerization and acetylation. Later, the kinetic study of oil adsorption in water of these fibers revealed that the untreated fiber adsorbed a total of 4.3 grams of oil per gram of fiber, reached saturation in 90 minutes, and exhibited a maximum adsorption of 3.8 grams at equilibrium. In contrast, the fibers treated with acetylation demonstrated a maximum adsorption capacity of 3.5 grams of oil per gram, reached saturation in just 30 minutes, and achieved a maximum adsorption of 4.4 grams. Meanwhile, the fiber treated with the ionic liquid also reached saturation in 30 minutes, with a maximum adsorption capacity of 3.5 grams of oil per gram and a maximum adsorption of 4.4 grams at equilibrium. These results indicate that the untreated fiber performs adequately in oil removal. Since it requires no treatment, the process is simpler and poses lower environmental risks. Consequently, the pristine fiber

shows promise as a potential biomaterial for removing organic pollutants from water, although further studies are needed to explore this prospect (Franca et al., 2019).
Other uses reported for this plant include:

- The crafting of flutes and guacharacas;
- The leaf midrib is used to weave fishing traps known as "guami";
- The inflorescence spike can be fashioned into arrows and harpoons for fishing;
- The stem is utilized to create items such as paper, chairs, beds, canoe paddles, cages, and cribs;
- Waste produced from scraping the midrib to extract fiber can be mixed with cow manure to create organic fertilizer;
- Leaf waste from the extraction of the central vein serves as feed for domestic animals like cows, horses, and donkeys (Casas Caro 2010; Jaimes Roncancio; 2023; Rojas Barbosa, 2003; Teisher and Stimmel, 2022).

4. Potential in the field of plant-fiber-based materials

As discussed in the previous section, *Gynerium sagittatum* shows significant potential, particularly due to its cellulose fibers and the various chemical modifications that can be applied to them. These modifications enable the adjustment of their physicochemical properties for specific applications. For instance, certain chemical alterations to cellulose have been proposed to reduce the number of hydroxyl groups by replacing them with functional groups that have a lower affinity for water (Kommula et al., 2014; Mora-Espinosa & Ramón-Valencia, 2017). This process reduces the hydrophilic character of the fibers, thereby enhancing the strength, rigidity, and durability of the resulting compounds. The literature suggests several alternatives for modifying cellulose, including alkalization, acetylation, and nitration; additionally, the use of ionic liquids can alter the surface of lignocellulosic material and facilitate further modifications (see **Figure 3**) (Kommula et al., 2014; Mora-Espinosa & Ramón-Valencia, 2017). These methods represent just a few of the numerous processes that can be performed on the fibers, as they are considered suitable substrates for layer-by-layer modifications and the development of composites. This makes it possible to expand the potential applications of these fibers in environmental remediation using polymeric supports that can be functionalized to impart selectivity for contaminants by incorporating molecules such as EDTA. In turn, this ability to modify the fibers is relevant to ongoing efforts to develop structural materials and panels, as well as their use in textiles, composite materials, packaging, emulsions, and bioactive wound dressings, among others.

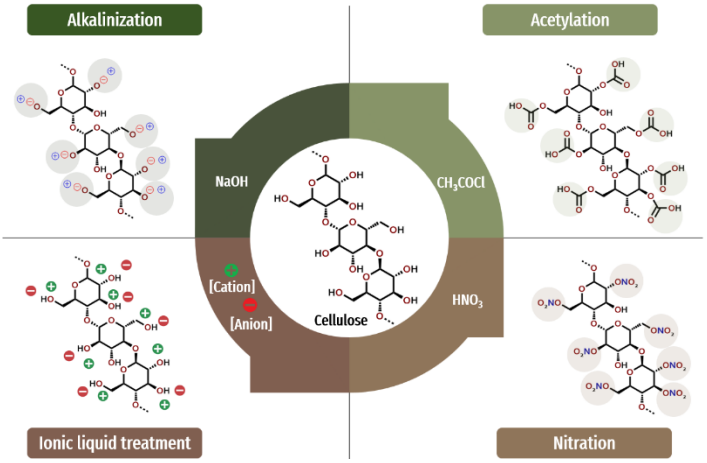


Figure 3. Modification of cellulose in *Gynerium sagittatum* fibers by different chemical strategies. Alkalinization, acetylation, and nitration modified the chemical structure of cellulose, while ionic liquid treatment altered intermolecular interactions and generated physical and structural changes.

4.1. Mercerization/Alkalinization

Mercerization, commonly referred to as alkalinization, is a process applied to natural fibers to enhance their surface and mechanical properties. This technique involves using concentrated solutions of mineral bases, such as sodium hydroxide, to disrupt the internal hydrogen bonds within the fiber and decrease the Van der Waals forces in the cellulose structure. As a result, the fiber's diameter may be reduced, and non-cellulosic components, such as lignin and hemicellulose, can be removed. Additionally, this process encourages the reorganization and separation of the fibers, which leads to increased surface roughness. This roughness is particularly beneficial for improving the interaction between the fibers and polymer matrices in reinforced composite applications (Chalid et al., 2015; Nabinejad et al., 2017; Verma & Goh, 2021).
Recent studies have indicated that the mercerization process can significantly alter the properties of arrow cane fibers. This process begins with a pretreatment using a boric acid-borax solution, followed by immersion in a 10 % (w/v) sodium hydroxide (NaOH) solution for 48 hours. After this treatment, the fibers are thoroughly washed with distilled water to remove any residual alkali and then oven-dried. Analysis of the fibers using Scanning Electron Microscopy (SEM) and Atomic Force Microscopy (AFM) revealed a marked reduction in surface levels of lignin, hemicellulose, waxes, and pectin. Additionally, the treatment resulted in a clearer definition of the microfiber structure, demonstrating the effectiveness of the mercerization process (Monsalve Alarcón et al., 2018).

4.2. Acetylation

Another alternative for modifying the surface properties of *Gynerium sagittatum* fiber - primarily to reduce its affinity for water

molecules - is acetylation. This process involves replacing hydroxyl groups with acetyl groups, which alters the hydrophilicity of the material and typically enhances the fiber's strength, durability, and moisture resistance. Nevertheless, some studies indicate that acetylated fibers may exhibit decreased thermal stability compared to the original, unmodified fibers; also, it has been observed that when acetylating arrow cane fibers, there is a change in the intensity of the peaks corresponding to the 2θ angles related to the crystalline and amorphous fractions of cellulose (Franca et al., 2019; Parveen et al., 2017; Wang et al., 2013).

To enhance the efficiency of the acetylation reaction, it is typically conducted after mercerization. This sequence helps reduce the presence of hemicellulose and lignin, increase the surface area of cellulose, and deprotonate the hydroxyl groups of the polysaccharide. This process can be monitored using mid-infrared spectroscopy, which shows a decrease in the intensity of the band associated with hydroxyl groups (-OH) and an increase in the signals associated with carbonyl groups (C=O) (Mokaloba & Batane, 2014; Tan et al., 2014; Franca et al., 2019). **Table 1** highlights the main vibrational frequencies used to monitor the chemical modifications in these fibers. Furthermore, SEM analysis has revealed that the combination of mercerization and acetylation leads to significant changes in fiber morphology, resulting in a transition from a rough to a smoother surface (Mokaloba & Batane, 2014; Tan et al., 2014).

4.3. Nitration

Nitration is a chemical modification process that introduces nitro groups (-NO_2) into the cellulose of fibers. This is achieved by concentrated nitric and sulfuric acids, which nitrate the hydroxyl groups in cellulose. This process leads to changes in the material's structural and functional properties. For example, nitration can reduce the fiber's absorption and swelling capacity (Costa et al., 2021).

Research has shown that incorporating nitro groups can promote a loss of cellulose crystallinity, decrease its mechanical properties, and increase its solubility in organic solvents. Additionally, the presence of nitro groups affects the material's electrostatic properties by enhancing interactions with cations. This characteristic is particularly valuable for cation exchange processes and the retention of harmful contaminants, such as heavy metal cations, which can pose risks to public health and the environment when present at levels exceeding permitted limits (Bolisetty et al., 2019; Li et al., 2016; Zamora-Ledezma et al., 2021).

Although the nitration of *Gynerium sagittatum* fibers has not been studied in the literature, it presents a significant research opportunity. In previous work, nitrated fibers for water remediation were produced from eucalyptus bark using a one-pot system that involved mixing the fibers with nitric acid. This mixture was heated to a temperature of 70 - 80 °C for 20 minutes. Afterward, the solution was allowed to cool to room temperature, then it was filtered and washed with distilled water to remove any residual acid and dyes (Zambrano et al., 2022).

Table 1. Relevant functional groups in the middle IR region that allow the monitoring of the chemical modifications of Arrow Cane fibers.

Functional group	Wavenumber (cm^{-1})	Reference
Hydroxyl -C-O-H	3600-3000	Franca et al., 2019 Xu et al., 2010
Alkyl C-H	~ 2935	Célino et al., 2014
Alkyl C-H_2	~ 2862	Célino et al., 2014
Carbonyl C=O	~ 1700	Célino et al; 2014 Franca et al., 2019
Aromatic rings (from lignin)	~1275 ~1595 ~1502	Célino et al; 2014

4.4. Ionic liquid treatment

On an industrial scale, several methods are employed to isolate cellulose from biomass, with the Organosolv and Kraft processes being the most notable. However, numerous studies have pointed out the challenges associated with these methods. In addition to degrading components such as lignin and hemicellulose, they necessitate expensive operating conditions, which include high temperatures, treatment of toxic effluents, and additional water purification processes (Isik et al., 2014; Sun et al., 2009). Consequently, research has focused on developing alternative methods, notably ionic liquids, which are salts in a liquid-salts that melt below 100 °C and remain liquid at room temperature; they exhibit high chemical stability and low vapor pressure and have proven capable of dissolving cellulose (Isik et al., 2014; Swatloski et al., 2002).

The earliest reports of this specific characteristic date back to the early 21st century, when cellulose was dissolved in an ionic liquid composed of 1-butyl-3-methylimidazolium cations and various anions. This process occurs because the anions in ionic liquids compete with the intermolecular and intramolecular hydrogen bonds in cellulose. As a result, these bonds break, leading to the formation of new electrostatic interactions that can be up to three times stronger than the original hydrogen bonds. Additionally, electrostatic interactions have been observed with the oxygen atoms of the hydroxyl (-OH) groups within the cellulose structure, which also aids in the dissolution of cellulose (Dong et al., 2016; Swatloski et al., 2002; Xu et al., 2010; Youngs et al., 2011).

Regarding research on the chemical modification of Arrow Cane fibers, the ionic liquid 2-hydroxyethylamine acetate has been reported as a material for removing oily contaminants in the aqueous phase. After using this ionic liquid, it was found that the fibers' crystallinity did not decrease significantly, indicating that this ionic liquid did not cause structural changes that would affect the cellulose matrix. Nevertheless, an evaluation of the oil sorption capacity in water revealed a reduction in sorption capacity in the treated fibers compared to the unmodified fibers. This suggests that

it may be necessary to explore other experimental conditions or to use different ionic liquids, such as [C_2mim][OAc] or [C_4mim][OAc], which have shown greater efficacy in modifying the crystalline structure of cellulose (Cheng et al., 2012; Franca et al., 2019; Zhang et al., 2014).

5. Conclusions and final remarks

Gynerium Sagittatum is a plant that thrives primarily in the tropical and subtropical regions of Latin America and Central America, where it has been recognized for its diverse therapeutic applications - ranging from treating alopecia to providing analgesic relief in kidney diseases - as well as its use in building houses and corrals, its ability to compact soil, and, most notably, the use of its leaf fiber to create handicrafts. While all this knowledge and these applications have largely ancestral origins and are based on experience, they often lack a comprehensive, clear, formal, and rigorous scientific understanding of the composition and physicochemical properties that enable these applications, particularly those not related to the plant's fiber.

Thus, there are situations in which the available phytochemical literature demonstrates that the species contains metabolites of interest. However, these studies do not provide a comprehensive characterization of the distribution of these compounds among the plant's different organs, nor do they address essential aspects such as age, cultivar, region, or growth conditions, or even a projection toward technological maturity through the optimization of extraction conditions.

This botanical aspect should be addressed to identify bioactive compounds that could serve as candidates for drug development through phytochemical and pharmacological screening, and to understand the biological pathways that enable the bioaccumulation of contaminants such as heavy metals, thereby establishing this plant as a potential bioaccumulator for phytoremediation in Latin America.

This technological maturity gap also exists in applications involving fibers and their byproducts, as few studies have explored ways to add value to them beyond handicrafts. However, despite this limitation, it has been shown that chemical and physical-structural modifications can modulate their properties, potentially enabling their use as a material for environmental remediation through surface interactions with contaminants. This highlights their potential as a biomaterial and substrate for chemical modification through the use of hydroxyl groups as a functionalization platform, an area of significant research interest for chemical, food, pharmaceutical, and materials applications.

Conflict interest. The authors declare that there is no conflict of interest.

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